

Class Activity-2

Task 1: Student Marks Record using CSV File

Code:

```
import csv
file = open("student_marks.csv", "w", newline="")
writer = csv.writer(file)
writer.writerow(["Register Number", "Student Name", "Department", "Python Marks", "Data Science Marks"])
for i in range(10):
    print("\nEnter details of Student", i + 1)
    regno = input("Register Number: ")
    name = input("Student Name: ")
    dept = input("Department: ")
    python = int(input("Python Marks: "))
    ds = int(input("Data Science Marks: "))
    writer.writerow([regno, name, dept, python, ds])
file.close()
print("\nData stored successfully in 'student_marks.csv'")
```

Output:

```
= RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/student_marks.py

Enter details of Student 1
Register Number: 382
Student Name: Ananya
Department: AI&DS
Python Marks: 77
Data Science Marks: 90

Enter details of Student 2
Register Number: 377
Student Name: Naga Deekshitha
Department: AI&ML
Python Marks: 80
Data Science Marks: 89

Enter details of Student 3
Register Number: 390
Student Name: Sophie
Department: CSE-AI
Python Marks: 90
Data Science Marks: 75

Enter details of Student 4
Register Number: 315
Student Name: Poornima
Department: AI&ML
Python Marks: 80
Data Science Marks: 88

Enter details of Student 5
Register Number: 317
Student Name: Sravanthi
Department: AI&ML
Python Marks: 90
Data Science Marks: 87

Enter details of Student 6
Register Number: 385
Student Name: Bhavya
Department: CSE
Python Marks: 87
Data Science Marks: 70
```

```
Enter details of Student 7
Register Number: 307
Student Name: Abhi
Department: CSE-AI
Python Marks: 67
Data Science Marks: 90

Enter details of Student 8
Register Number: 317
Student Name: Arjun
Department: AI&DS
Python Marks: 77
Data Science Marks: 87

Enter details of Student 9
Register Number: 415
Student Name: Gowtami
Department: CSE
Python Marks: 88
Data Science Marks: 87

Enter details of Student 10
Register Number: 232
Student Name: Yash
Department: CSE-AI
Python Marks: 86
Data Science Marks: 73

Data stored successfully in 'student_marks.csv'
```

Task 2: Employee Database using CSV File

Code:

```
import csv
file = open("employees.csv", "w", newline="")
writer = csv.writer(file)
writer.writerow(["Employee ID", "Employee Name", "Department", "Designation", "Salary"])
for i in range(5):
    print("\nEnter details of Employee", i + 1)
    emp_id = input("Employee ID: ")
    name = input("Employee Name: ")
    dept = input("Department: ")
    desig = input("Designation: ")
    salary = input("Salary: ")
    writer.writerow([emp_id, name, dept, desig, salary])
file.close()
print("\nEmployee details stored successfully in 'employees.csv'")
```

Output:

```
>>>
===== RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/employee_details.py =====
Enter details of Employee 1
Employee ID: 777
Employee Name: Ananya
Department: AI&DS
Designation: Data Analyst
Salary: 70000
Enter details of Employee 2
Employee ID: 222
Employee Name: Naga Deekshitha
Department: AI&ML
Designation: Software Engineer
Salary: 60000
Enter details of Employee 3
Employee ID: 191
Employee Name: Swarna
Department: CSE-AI
Designation: AI Engineer
Salary: 80000
Enter details of Employee 4
Employee ID: 181
Employee Name: Bhavya
Department: CSE
Designation: Software Engineer
Salary: 50000
Enter details of Employee 5
Employee ID: 315
Employee Name: Sravanthi
Department: AI&ML
Designation: AI Engineer
Salary: 75000
Employee details stored successfully in 'employees.csv'
>>>
```

Task 3: Faculty Information System using JSON File

Code:

```
import json
faculty_list = []
# Input details of 5 faculty members
for i in range(5):
    print("\nEnter details of Faculty", i + 1)
    faculty = {
        "Faculty ID": input("Faculty ID: "),
        "Faculty Name": input("Faculty Name: "),
        "Department": input("Department: "),
        "Designation": input("Designation: "),
        "Experience (Years)": int(input("Experience (Years): "))
    }
    faculty_list.append(faculty)
# Write to JSON file
with open("faculty.json", "w") as file:
    json.dump(faculty_list, file, indent=4)
print("\nFaculty details stored successfully in 'faculty.json'")
```

Output:

```

===== RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/faculty information.py =====
Enter details of Faculty 1
Faculty ID: 101
Faculty Name: Arjun
Department: AI&DS
Designation: Professor
Experience (Years): 7

Enter details of Faculty 2
Faculty ID: 102
Faculty Name: Sarala
Department: CSE
Designation: Associate Professor
Experience (Years): 10

Enter details of Faculty 3
Faculty ID: 103
Faculty Name: David
Department: AI&ML
Designation: Associate Professor
Experience (Years): 17

Enter details of Faculty 4
Faculty ID: 104
Faculty Name: Krunal
Department: CSE-AI
Designation: Professor
Experience (Years): 5

Enter details of Faculty 5
Faculty ID: 105
Faculty Name: Vidya
Department: Cyber
Designation: Professor
Experience (Years): 4

Faculty details stored successfully in 'faculty.json'
>>>

```

Task 4: Product Inventory Management using JSON File

Code:

```

import json
products = []
# Input details of 10 products
for i in range(10):
    print("\nEnter details of Product", i + 1)
    product = {
        "Product ID": input("Product ID: "),
        "Product Name": input("Product Name: "),
        "Category": input("Category: "),
        "Price": float(input("Price: ")),
        "Quantity Available": int(input("Quantity Available: "))
    }
    products.append(product)
# Write data to JSON file
with open("products.json", "w") as file:
    json.dump(products, file, indent=4)
print("\nProduct details stored successfully in 'products.json'")

```

Output:

== RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/product.py ==

Enter details of Product 1

Product ID: P101

Product Name: Laptop

Category: Electronic

Price: 70000

Quantity Available: 7

Enter details of Product 2

Product ID: P102

Product Name: Mobile

Category: Electronic

Price: 40000

Quantity Available: 10

Enter details of Product 3

Product ID: P103

Product Name: Lap top Charger

Category: Electronic

Price: 1000

Quantity Available: 5

Enter details of Product 4

Product ID: P104

Product Name: Monitor

Category: Electronic

Price: 12000

Quantity Available: 20

Enter details of Product 5

Product ID: P105

Product Name: SmartPhone Charger

Category: Electronic

Price: 1500

Quantity Available: 20

Enter details of Product 6

Product ID: P106

Product Name: EarPhones

Category: Electronic

Price: 1000

Quantity Available: 30

```
Enter details of Product 7
Product ID: P107
Product Name: Pendrive
Category: Electronic
Price: 700
Quantity Available: 40

Enter details of Product 8
Product ID: P108
Product Name: Webcam
Category: Electronic
Price: 3000
Quantity Available: 26

Enter details of Product 9
Product ID: P109
Product Name: Mouse
Category: Electronic
Price: 1000
Quantity Available: 30

Enter details of Product 10
Product ID: P110
Product Name: Printer
Category: Electronic
Price: 5000
Quantity Available: 32

Product details stored successfully in 'products.json'
```

Task 5: Course Information Database using XML File

Code:

```
import xml.etree.ElementTree as ET
# Root element
root = ET.Element("Courses")
# Input details of 5 courses
for i in range(5):
    print("\nEnter details of Course", i + 1)
    course = ET.SubElement(root, "Course")
    ET.SubElement(course, "CourseCode").text = input("Course Code: ")
    ET.SubElement(course, "CourseName").text = input("Course Name: ")
    ET.SubElement(course, "Credits").text = input("Credits: ")
    ET.SubElement(course, "Department").text = input("Department: ")
    ET.SubElement(course, "FacultyName").text = input("Faculty Name: ")
# Create XML tree and write to file
```

```
tree = ET.ElementTree(root)
tree.write("courses.xml")
print("\nCourse details stored successfully in 'courses.xml'")
```

Output:

```
= RESTART: C:\Users\uppal\AppData\Local\Programs\Python\Python313\course information.py

Enter details of Course 1
Course Code: CSA101
Course Name: Python Programming
Credits: 4
Department: CSE
Faculty Name: Dr.Kumar

Enter details of Course 2
Course Code: AI201
Course Name: Machine Learning
Credits: 4
Department: AI&DS
Faculty Name: Dr.Priya

Enter details of Course 3
Course Code: DS307
Course Name: Data Science
Credits: 3
Department: AI&DS
Faculty Name: Dr.Rahul

Enter details of Course 4
Course Code: EC205
Course Name: Digital Electronics
Credits: 2
Department: ECE
Faculty Name: Dr.Meena

Enter details of Course 5
Course Code: IT210
Course Name: Database Management System
Credits: 3
Department: IT
Faculty Name: Dr.Suresh

Course details stored successfully in 'courses.xml'
```

Task 6: Hospital Patient Management System using XML File

Code:

```
import xml.etree.ElementTree as ET
# Create root element
root = ET.Element("Patients")
# Input details of 10 patients
for i in range(10):
    print("\nEnter details of Patient", i + 1)
```

```

patient = ET.SubElement(root, "Patient")
ET.SubElement(patient, "PatientID").text = input("Patient ID: ")
ET.SubElement(patient, "PatientName").text = input("Patient Name: ")
ET.SubElement(patient, "Age").text = input("Age: ")
ET.SubElement(patient, "Disease").text = input("Disease: ")
ET.SubElement(patient, "DoctorAssigned").text = input("Doctor Assigned: ")
# Save XML file
tree = ET.ElementTree(root)
tree.write("patients.xml")
print("\nPatient details stored successfully in 'patients.xml'")

```

Output:

```

===== RESTART: C:/Users/uppal/AppData/Local/Programs/Python/Python313/hospital_patient.py

Enter details of Patient 1
Patient ID: P001
Patient Name: Rahul
Age: 35
Disease: Fever
Doctor Assigned: Dr.Kumar

Enter details of Patient 2
Patient ID: P002
Patient Name: Priya
Age: 28
Disease: Diabetes
Doctor Assigned: Dr.Meena

Enter details of Patient 3
Patient ID: P003
Patient Name: Arjun
Age: 45
Disease: Hypertension
Doctor Assigned: Dr.Suresh

Enter details of Patient 4
Patient ID: P004
Patient Name: Kavya
Age: 30
Disease: Asthma
Doctor Assigned: Dr.Priya

Enter details of Patient 5
Patient ID: P005
Patient Name: Sneha
Age: 24
Disease: Migraine
Doctor Assigned: Dr.Meena

Enter details of Patient 6
Patient ID: P006
Patient Name: Vijay
Age: 40
Disease: COVID-19
Doctor Assigned: Dr.Rahul

```


Enter details of Patient 7
Patient ID: P007
Patient Name: Ramesh
Age: 40
Disease: Arthritis
Doctor Assigned: Dr.Kumar

Enter details of Patient 8
Patient ID: P008
Patient Name: Anitha
Age: 33
Disease: Dengue
Doctor Assigned: Dr.Suresh

Enter details of Patient 9
Patient ID: P009
Patient Name: Kiran
Age: 60
Disease: Heart Disease
Doctor Assigned: Dr.Kumar

Enter details of Patient 10
Patient ID: P010
Patient Name: Divya
Age: 26
Disease: Typhoid
Doctor Assigned: Dr.Priya

Patient details stored successfully in 'patients.xml'
